

# Advanced Kernel methods

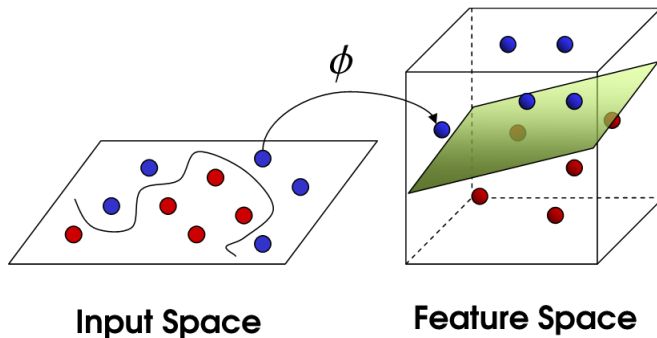
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Hubert Curien laboratory

# Quiz

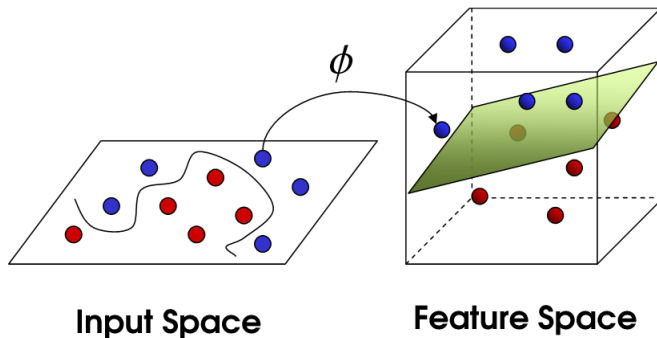
- Q1) What is the difference between classification and regression problems ?
- Q2) Can you give some examples of methods/algorithms that can be used for a regression problem ?
- Q3) What do you think kernel methods are/do ?

- Motivation for using kernel methods.
- The (kernel) ridge regression example.
- The representer theorem.
- Benefits and drawbacks of kernel methods.
- Nystöm approximation to speed up kernel methods.

# Motivation in classification problem



# Motivation in classification problem



$\Phi$  is often unknown !  
but  $k(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle$  is known

# Notations

For  $\mathcal{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}$  a set of  $n$  data points, where  $x_i \in \mathbb{R}^d$  and  $y_i \in \mathbb{R}$ . We denote by:

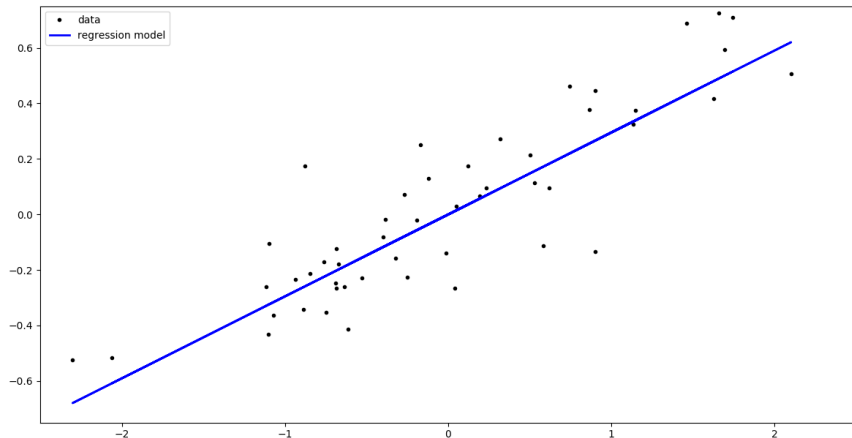
- $X$  the data matrix:

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^{n \times d}$$

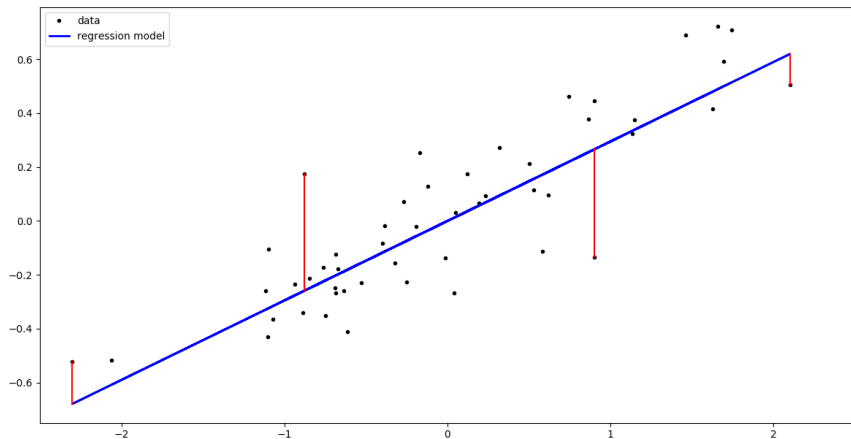
- $y$  the labels vector:

$$y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n$$

# Linear regression problem



# Linear regression problem





# Linear ridge regression

Let  $\mathcal{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}$  be a set of  $n$  data points:

$$w^* = \arg \min_{w \in \mathbb{R}^d} \sum_{i=1}^n (y_i - \langle w, x_i \rangle)^2 + \lambda \|w\|_2^2$$

# Linear ridge regression

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Can you write it in matrix form ?

# Linear ridge regression using matrix notations

Since:

$$Xw = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} w_1, w_2, \dots, w_d \end{bmatrix} = \begin{bmatrix} \langle x_1, w \rangle \\ \langle x_2, w \rangle \\ \vdots \\ \langle x_n, w \rangle \end{bmatrix}$$

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We have:

$$\|y - Xw\|_2^2 = \left\| \begin{bmatrix} y_1 - \langle x_1, w \rangle \\ y_2 - \langle x_2, w \rangle \\ \vdots \\ y_n - \langle x_n, w \rangle \end{bmatrix} \right\|_2^2 = \sum_{i=1}^n (y_i - \langle w, x_i \rangle)^2$$

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and thus:

$$w^* = \arg \min_{w \in \mathbb{R}^d} \sum_{i=1}^n (y_i - \langle w, x_i \rangle)^2 + \lambda \|w\|_2^2 = \arg \min_{w \in \mathbb{R}^d} \|y - Xw\|_2^2 + \lambda \|w\|_2^2$$

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Can you find the optimum solution ?

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Can you find the optimum solution ?

1) compute the gradient of

$$L(w) = \frac{1}{2} \|y - Xw\|_2^2 + \lambda \|w\|_2^2 = \frac{1}{2} w^T X^T X w - w^T X^T y + \frac{1}{2} y^T y + \lambda w^T w$$



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$$w^* = (X^T X + \lambda I)^{-1} X^T y$$

How can we "kernelize" the ridge regression method ?

# Kernel ridge regression

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Replacing  $x$  by  $\Phi(x)$  in the optimization problem ?

# Kernel ridge regression

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$$\begin{aligned} w^* &= \arg \min_{w \in \mathbb{R}^d} \sum_{i=1}^n (y_i - \langle w, \Phi(x_i) \rangle)^2 + \lambda \|w\|_2^2 \\ &= \arg \min_{w \in \mathbb{R}^d} \|y - \Phi w\|_2^2 + \lambda \|w\|_2^2 \end{aligned}$$

where:

$$\Phi = \begin{bmatrix} \Phi(x_1) \\ \vdots \\ \Phi(x_n) \end{bmatrix}$$

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What is the optimal solution in this case ?

$$w^* = \left( \Phi^T \Phi + \lambda I \right)^{-1} \Phi^T y$$



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where:

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What is the optimal solution in this case ?

$$w^* = \left( \Phi^T \Phi + \lambda I \right)^{-1} \Phi^T y$$

Can't be used because  $\Phi$  is unknown !

# Kernel ridge regression

Replace the problem:

$$w^* = \arg \min_{w \in \mathbb{R}^d} \sum_{i=1}^n (y_i - \langle w, \Phi(x_i) \rangle)^2 + \lambda \|w\|_2^2$$

by:

$$f^* = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{F}}^2$$

where  $\mathcal{F}$  is an infinite set of functions.

# Positive definite kernels

## positive definite (p.d.) kernel definition

Let  $\mathcal{X}$  be a non-empty set. A symmetric function  $k : \mathcal{X} \times \mathcal{X} \Rightarrow \mathbb{R}$  is called a positive-definite (p.d.) kernel on  $\mathcal{X}$  if:

$$\sum_{i=1}^n \sum_{j=1}^n c_i c_j k(x_i, x_j) \geq 0,$$

holds for any  $\{x_i\}_{i=1}^n \in \mathcal{X}$  and  $\{c_i\}_{i=1}^n \in \mathbb{R}$ .

## RKHS definition

Let  $\mathcal{X}$  be a non-empty set, and  $\mathcal{H} = \{f | f : \mathcal{X} \Rightarrow \mathbb{R}\}$  an Hilbert space of functions endowed with the dot product  $\langle ., . \rangle$ .

$\mathcal{H}$  is a reproducing kernel Hilbert space if there exists a p.d. kernel  $k$  such that:

- $\forall x \in \mathcal{X}, \forall f \in \mathcal{H}: f(x) = \langle f, k(x, .) \rangle_{\mathcal{H}} = \langle f, \phi(x) \rangle_{\mathcal{H}}$
- $\mathcal{H} = \text{span}(\{k(x, .)\}_{x \in \mathcal{X}})$

# The representer theorem

## The representer theorem

- Let  $\mathcal{X}$  be a nonempty set, a p.d. kernel  $k$ ,  $\mathcal{H}$  the corresponding RKHS,
- Let  $\mathcal{D} = \{x_1, \dots, x_n\} \subseteq \mathcal{X}$  be a finite set of  $n$  points in  $\mathcal{X}$ ,
- Let  $g$  be a strictly increasing real-valued function,
- Let  $L$  be an arbitrary error function.

Then any solution to the optimization problem:

$$f^* = \arg \min_{f \in \mathcal{H}} L((x_1, y_1, f(x_1)), \dots, (x_n, y_n, f(x_n))) + g(\|f\|),$$

admit a representation of the form:

$$f^*(.) = \sum_{i=1}^n \alpha_i k(., x_i) \quad \text{with } \alpha_i \in \mathbb{R}$$

# Proof of the representer theorem

Let decompose:  $f = \sum_{i=1}^n a_i \phi(x_i) + v$  with  $\langle v, \phi(x_i) \rangle = 0 \ \forall \ i$ .

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We thus have:  $f(x_j) = \langle f, k(x_j, \cdot) \rangle_{\mathcal{H}} = \langle f, \phi(x_j) \rangle_{\mathcal{H}}$

$$\begin{aligned} &= \left\langle \sum_{i=1}^n a_i \phi(x_i) + v, \phi(x_j) \right\rangle_{\mathcal{H}}, \\ &= \left\langle \sum_{i=1}^n a_i \phi(x_i), \phi(x_j) \right\rangle + \cancel{\langle v, \phi(x_j) \rangle} \overset{0}{\rightarrow}. \end{aligned}$$

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For the second term:

$$\begin{aligned} g(\|f\|) &= g\left(\left\| \sum_{i=1}^n a_i \phi(x_i) + v \right\|\right) = g\left(\sqrt{\left\| \sum_{i=1}^n a_i \phi(x_i) \right\|^2 + \|v\|^2}\right) \\ &\geq g\left(\sqrt{\left\| \sum_{i=1}^n a_i \phi(x_i) \right\|^2}\right) \end{aligned}$$



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$\implies$  setting  $v = 0$  doesn't affect the optim. pb!

# Back to kernel Ridge Regression

Reminder: the KRR optimization problem is:

$$f^* = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{F}}^2$$

with:

- $g(\|f\|) = \lambda \|f\|_{\mathcal{F}}^2$
- $L((x_1, y_1, f(x_1)), \dots, (x_n, y_n, f(x_n))) = \sum_{i=1}^n (y_i - f(x_i))^2$

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Then, using the representer theorem we have:

$$f^*(.) = \sum_{j=1}^n \alpha_j k(., x_j) \quad \text{with } \alpha_i \in \mathbb{R}$$

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we can rewrite the KRR problem in terms of  $\alpha \in \mathbb{R}^n$ .

# Back to kernel Ridge Regression

The KRR optimization problem becomes:

$$f^* = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \|f\|_{\mathcal{F}}^2$$

$$\begin{aligned} \alpha^* &= \arg \min_{\alpha \in \mathbb{R}^n} \sum_{i=1}^n \left( y_i - \sum_{j=1}^n \alpha_j k(x_i, x_j) \right)^2 + \lambda \left\| \sum_{j=1}^n \alpha_j k(., x_j) \right\|_{\mathcal{F}}^2 \\ &= \arg \min_{\alpha \in \mathbb{R}^n} \|y - K\alpha\|_2^2 + \lambda \alpha^T K \alpha \end{aligned}$$

where

$$K = \Phi \Phi^T = \begin{bmatrix} \Phi(x_1) \\ \Phi(x_2) \\ \vdots \\ \Phi(x_n) \end{bmatrix} \begin{bmatrix} \Phi(x_1), \Phi(x_2), \dots, \Phi(x_n) \end{bmatrix}$$

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Convex optimization problem !

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What is the solution in this case ?

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Convex optimization problem !

What is the solution in this case ?

Since:

$$\frac{\partial L}{\partial \alpha} = KK\alpha - Ky^T + \lambda K\alpha = K(K\alpha - y^T + \lambda\alpha) = 0$$

means:

$$(K\alpha - y^T + \lambda\alpha) = 0$$

we thus have:

$$\alpha^* = (K + \lambda I)^{-1} y^T.$$



# Back to kernel Ridge Regression

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means:

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we thus have:

$$\alpha^* = (K + \lambda I)^{-1} y^T.$$

We can use it since  $K$  is known !

# Benefits and drawbacks of kernel methods

## Benefits:

- Can work on small size datasets, unlike deep learning methods.
- More interpretable than deep learning methods.
- Well established theory.

## Drawbacks:

- They are slow (quadratic or cubic time) and do not scale with large datasets.
- They do not achieve the same level of performance as deep learning methods.

The Nyström method.

# The Gram matrix

## Gram matrix

Given a kernel function  $k$  and a set of data  $\mathcal{D} = \{x_i\}_{i=1}^n$ , the  $n \times n$  Gram matrix is defined as:

$$K = \Phi \Phi^T = \begin{bmatrix} \Phi(x_1) \\ \Phi(x_2) \\ \vdots \\ \Phi(x_n) \end{bmatrix} \begin{bmatrix} \Phi(x_1), \Phi(x_2), \dots, \Phi(x_n) \end{bmatrix}$$

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Computing the Gram matrix is done in:  $\Theta(n^2)$

Computing its inverse is done in:  $\Theta(n^3)$

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Computing the Gram matrix is done in:  $\Theta(n^2)$

Computing its inverse is done in:  $\Theta(n^3)$

Too long when  $n$  is large !

# The complexity of KRR

**Reminder:** the solution of KRR is:  $\alpha^* = (K + \lambda I)^{-1} y^T$

**Remark 1:** computing  $(K + \lambda I)^{-1}$  is done in  $\Theta(n^3)$ .

# The complexity of KRR

**Reminder:** the solution of KRR is:  $\alpha^* = (K + \lambda I)^{-1} y^T$

**Remark 1:** computing  $(K + \lambda I)^{-1}$  is done in  $\Theta(n^3)$ .

**Remark 2:** if  $K = ABA^T$  such that  $A \in \mathbb{R}^{n \times s}$ ,  $B \in \mathbb{R}^{s \times s}$  with  $s \ll n$ , then using the Woodberry equality:

$$(K + \lambda I)^{-1} = \lambda^{-1} I + A \left( B^{-1} + \lambda^{-1} A^T A \right)^{-1} A^T$$

is computed in  $\Theta(s^3 + ns^2) < \Theta(n^3)$  when  $s < n$ .



# The Nyström method

## Definition: Nyström method (Williams and Seeger, 2000)

Let  $k$  be a kernel function,  $\mathcal{D} = \{x_i\}_{i=1}^n$  a set of  $n$  data points, and  $\mathcal{S} \subseteq \mathcal{D}$  a subset of points.

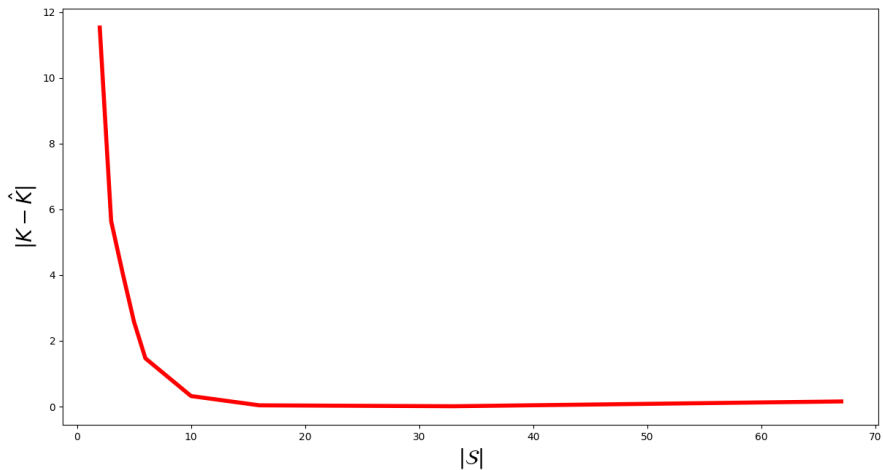
The matrix:

$$\hat{K} = K_{n,s} K_{s,s}^+ K_{n,s}^T$$

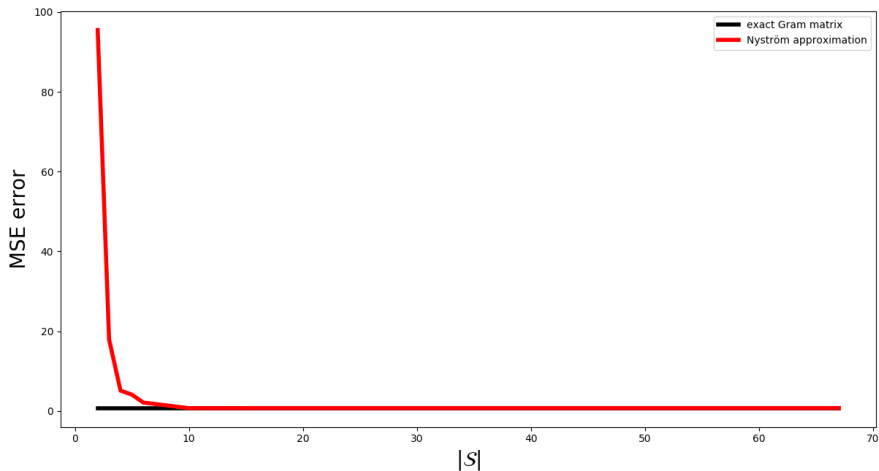
is a low rank approximation of the Gram matrix  $K$ , where:

- $K_{n,s} = [k(x, x')]_{x \in \mathcal{D}, x' \in \mathcal{S}} \in \mathbb{R}^{n \times |\mathcal{S}|}$ ,
- $K_{s,s} = [k(x, x')]_{x \in \mathcal{S}, x' \in \mathcal{S}} \in \mathbb{R}^{|\mathcal{S}| \times |\mathcal{S}|}$ .

# Nyström: some experiments



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# Conclusion

- The Nyström method reduces the complexity of some 'kernelized' ML algorithms.
- The Nyström method can be used in several methods: SVM, PCA, distance between probabilities ...
- The quality of its prediction depends on the quality of  $\mathcal{S}$ .
- May provide better predictions than the exact method in some cases !

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How to choose a good subset  $\mathcal{S}$  ?

Answer: a whole field of research...

So, do you like kernel methods now ?